

CHEMICAL KINETICS

Chemical Kinetics is branch of chemistry that deals with the rates of reaction or factors affecting rate and the mechanism of the reaction.

* RATE OF REACTION

The rate of chemical reaction is defined as the change in concentration of a designated species i.e. reactant or product in the given time interval.

* AVERAGE RATE

It is defined as change in concentration of reactant or product in a given interval of time.

Consider a reaction



$$\text{Avg. rate of Rxn} = \frac{\Delta [C]}{\Delta t}$$

$$\therefore \text{Avg. rate of consumption of R} = \frac{-\Delta [R]}{\Delta t}$$

Because rate cannot be negative

$$\therefore \text{Avg. rate of formation of P} = \frac{\Delta [P]}{\Delta t}$$

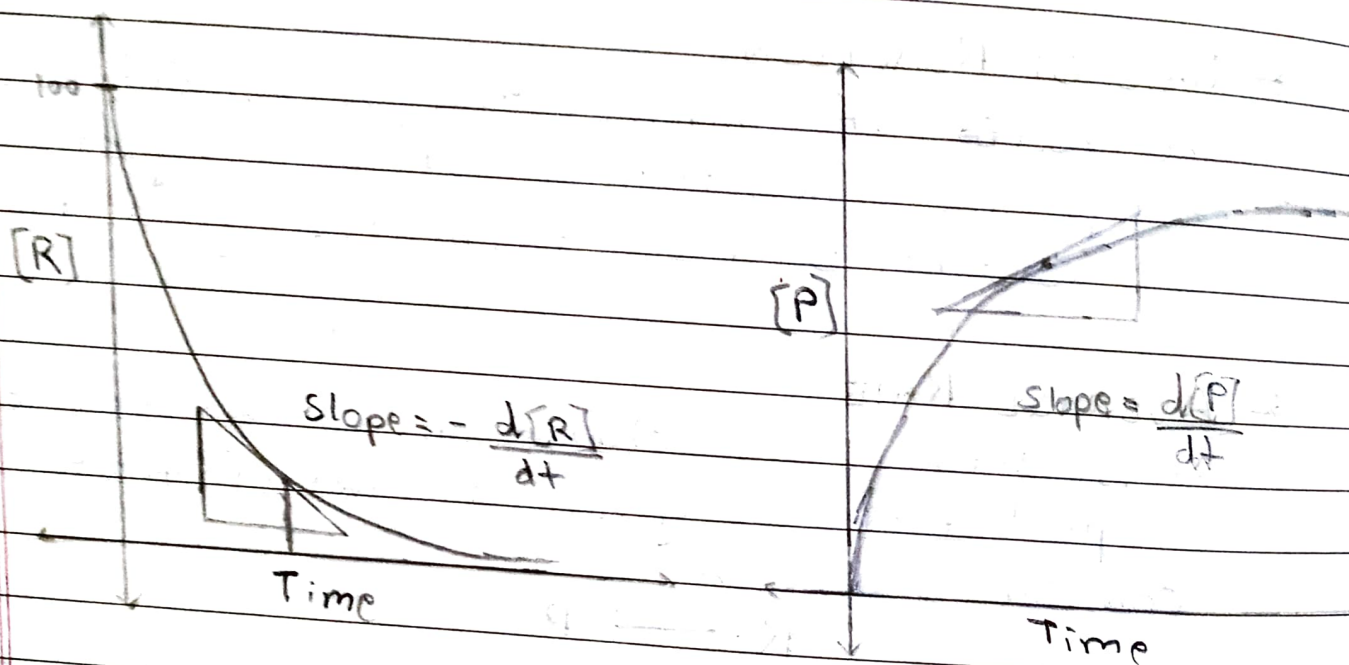
$$\therefore \boxed{\frac{\Delta [C]}{\Delta t} = \frac{-\Delta [R]}{\Delta t} = \frac{\Delta P}{\Delta t}}$$

$$\begin{aligned} \text{Unit} &= \text{M/s} \\ &= \text{mol/dm}^3 \text{s} \end{aligned}$$

Instantaneous Rate

The rate of reaction at a given instant of time is called as instantaneous rate of reaction.

To determine instantaneous rate of reaction, a graph of concentration versus time is plotted. A curve is obtained and the tangent is drawn at a point on a curve where the slope is rate of reaction.



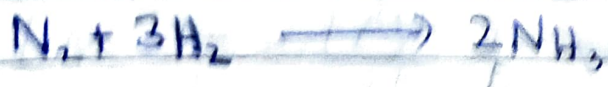
Instantaneous rate of consumption of $R = -\frac{d[R]}{dt}$

Instantaneous rate of formation of $P = \frac{d[P]}{dt}$

$\therefore$ Instantaneous rate of Reaction $= -\frac{d[R]}{dt} = \frac{d[P]}{dt}$

$$\begin{aligned}\text{Unit} &= \text{M s}^{-1} \\ &= \text{mol dm}^{-3} \text{s}^{-1}\end{aligned}$$

For Haber's Process



$$\text{Rate of consumption of } \text{N}_2 = -\frac{d[\text{N}_2]}{dt}$$

$$\text{Rate of consumption of } \text{H}_2 = -\frac{d[\text{H}_2]}{dt}$$

$$\text{Rate of formation of } \text{NH}_3 = \frac{d[\text{NH}_3]}{dt}$$

$$\begin{aligned} \text{Rate of consumption of } \text{H}_2 \\ = 3 \times \text{Rate of consumption of } \text{N}_2 \end{aligned}$$

$$-\frac{d[\text{H}_2]}{dt} = 3 \left(-\frac{d[\text{N}_2]}{dt} \right)$$

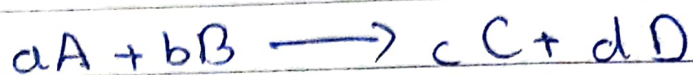
$$-\frac{1}{3} \frac{d[\text{H}_2]}{dt} = -\frac{d[\text{N}_2]}{dt} \quad \text{--- (1)}$$

$$\begin{aligned} \text{Rate of formation of } \text{NH}_3 \\ = 2 \times \text{Rate of consumption of } \text{N}_2 \end{aligned}$$

$$\frac{1}{2} \frac{d[\text{NH}_3]}{dt} = -\frac{d[\text{N}_2]}{dt} \quad \text{--- (2)}$$

$\therefore$ From (1) and (2)

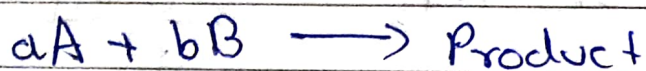
$$\text{Rate of Reaction} = -\frac{d[\text{N}_2]}{dt} = -\frac{1}{3} \frac{d[\text{H}_2]}{dt} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt}$$



$$\text{Rate of Reaction} = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$$

* Rate law and Rate constant

Consider a general reaction



The differential rate law is given as

$$\text{Rate} = \frac{1}{a} \frac{d[A]}{dt} \propto [A]^x [B]^y$$

$$\text{Rate} \propto [A]^x [B]^y$$

The rate law is defined as the experimentally determined equation where rate ^{is directly proportional to} $\propto$ concentration of reactants raise to some simple power.

Here x and y are called order of reaction and it can be positive, negative, or even zero or fraction.

$$\text{Rate} = k [A]^x [B]^y$$

where k is rate constant or specific rate constant and it is defined as the rate of reaction when the concentration terms are set to unity.

* ORDER OF REACTION

Consider a general reaction



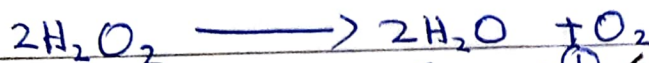
If the rate of reaction is

$$\text{Rate} = k [A]^x [B]^y$$

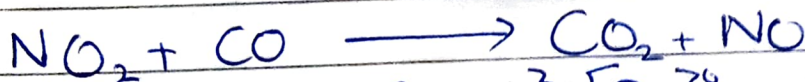
Experimentally determined

Then the sum of $x+y$ gives overall order of reaction. Thus, overall order of reaction is given as the sum of power of the concentration terms in the rate law expression.

Example



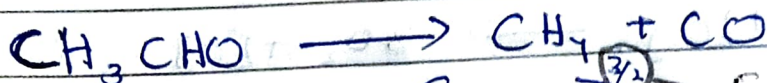
$$\text{Rate} = k [H_2O_2]^1 \longleftarrow 1^{\text{st}} \text{ order}$$



$$\text{Rate} = k [NO_2]^2 [CO]^0$$

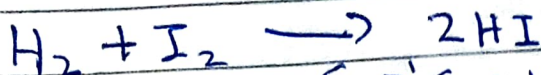
$\uparrow$
2nd Order
w.r.t NO_2

$\uparrow$
Zero w.r.t CO



$$\text{Rate} = [CH_3CHO]^{3/2}$$

Experimentally determined value



$$\text{Rate} = k [H_2]^1 [I_2]^1$$

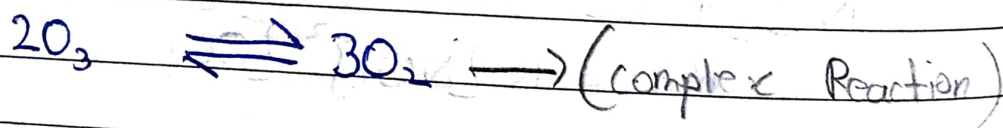
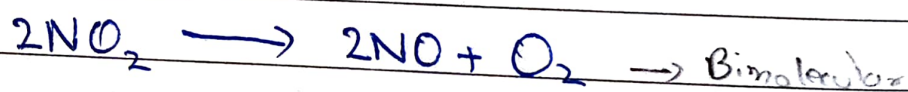
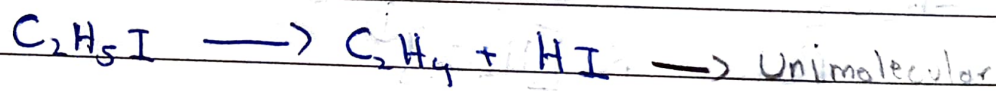
* Molecularity of elementary reaction

Complex Reaction: Complex Reactions are those reactions which takes place in a series of steps where each step is Elementary Reaction.

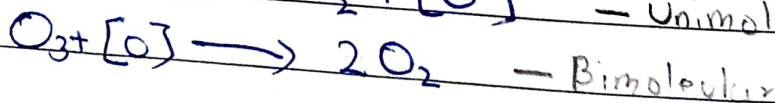
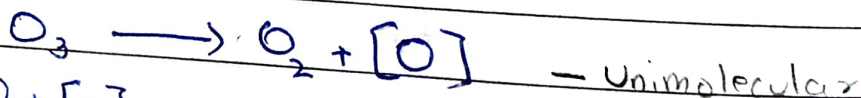
Elementary Reaction: Reactions which occur in a single step and cannot be broken down further into simpler reaction are called elementary reactions.

Molecularity: It is a number of reactant molecules taking part in an Elementary reaction.

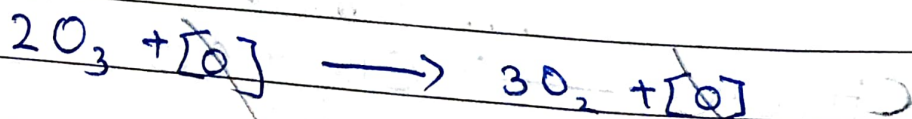
Example



Mechanism

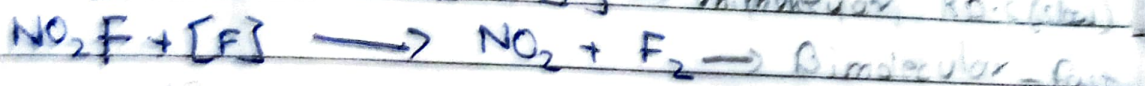


Overall
rxn





Mechanism:



Here F is reaction intermediate. The first step in the mechanism is the slowest step. It is Rate determining Step.

Reaction Intermediate

During reaction mechanism, some species are formed in one step and are consumed in the subsequent step. Such species are known as reaction intermediate. The concentration of reaction intermediate is difficult to determine and therefore they do not appear in rate law.

Rate determining Step

A complex reaction takes place in a series of elementary steps. One of these steps is slower than others and it is referred to as rate determining step.

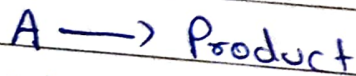
Molecularity of a reaction can be the order of reaction however, ~~order of reaction~~ converse is not true. Since, the rate determining step is an elementary reaction, the stoichiometry of the reaction is equal to the order of reaction however, order is not equal to molecularity.

ORDER	Molecularity
It is an experimentally determined property	It is a theoretical property
It is the sum of the powers of a concentration terms of reactants those appear in the rate law	It is the number of reactant molecules taking part in a elementary reaction.
It may be an integer, fraction or zero	It is a positive integer

* INTEGRATED RATE LAW

Integrated Rate law for 1st order reaction is defined as the rate of reaction when the depends on the concentration of the single reactant.

Consider a Reaction



∴ The differential rate law

$$\text{Rate} = -\frac{d[A]}{dt} = k[A]^1$$

$$-\frac{d[A]}{dt} = k[A]$$

$$-\frac{d[A]}{[A]} = k dt$$

On Integration

$$-\int_{[A_0]}^{[A_t]} \frac{d[A]}{[A]} = \int_{t=0}^{t=t} k dt$$

$$-\left\{ \ln [A] \right\}_{[A_0]}^{[A_t]} = k [t]_0^t$$

$$-\left\{ \ln [A_t] - \ln [A_0] \right\} = k(t-0) \quad \text{--- ①}$$

$$\ln [A_0] - \ln [A_t] = kt$$

$$\frac{\ln [A_0]}{[A_0]} = -kt$$

$$\frac{[A_t]}{[A_0]} = e^{-kt}$$

$$\therefore [A_t] = [A_0] e^{-kt} \quad \text{--- I}^{\text{st}} \text{ form}$$

From equation ①

$$-\ln [A_t] + \ln [A_0] = kt$$

$$\ln \frac{[A_0]}{[A_t]} = kt$$

$$2.303 \log_{10} \frac{[A_0]}{[A_t]} = kt$$

$$\therefore \boxed{k = \frac{2.303}{t} \log \frac{[A_0]}{[A_t]}} \quad \text{---} \star$$

$\therefore [A_0] = a =$ Initial concentration of A
and x is the amount that undergoes
reaction then concentration at time $t = [A_t] = a - x$

$$k = \frac{2.303}{t} \log \frac{a}{(a-x)}$$

Unit : s^{-1}

* HALF LIFE OF FIRST ORDER REACTION

It is the time required for the reactant concentration to fall to one half of its initial concentration.
The Integrated rate law for 1st order reaction is

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A_t]}$$

Where $[A_0]$ is initial concentration of reactant at $t=0$ and $[A_t]$ is concentration at time ' t '.
 $\therefore$ the time required for $[A_0]$ to become $[A_0]/2$ is given by $t_{1/2}$

Substituting in above equation we get

If $t = t_{1/2}$ $[A_t] = \frac{[A_0]}{2}$

$$K = \frac{2.303}{t_{1/2}} \log \frac{[A_0]}{[A_0]/2}$$

$$= \frac{2.303}{t_{1/2}} \log 2$$

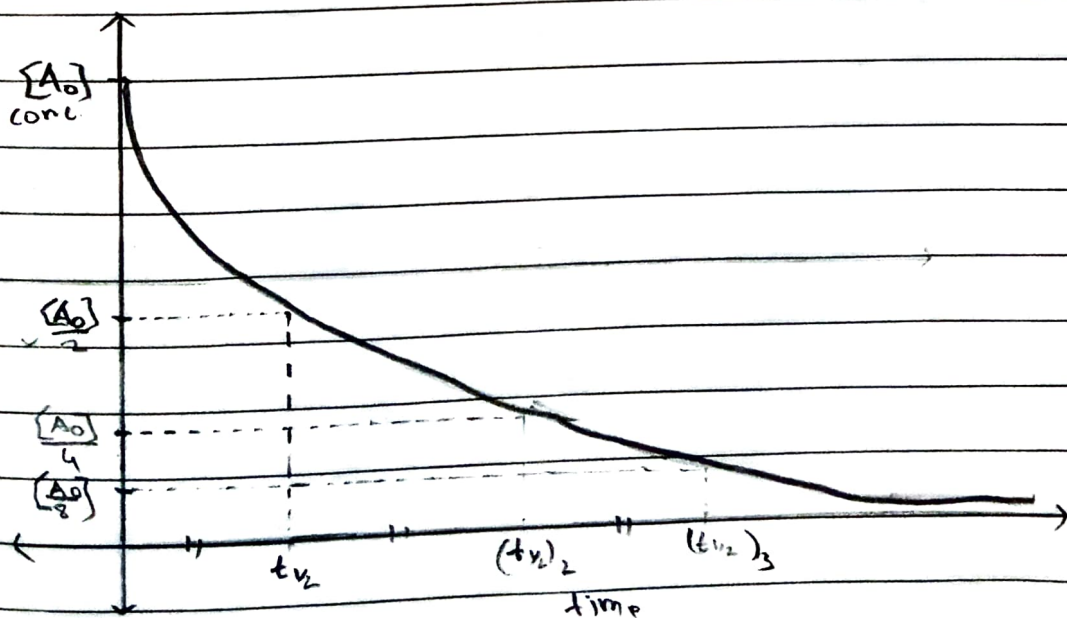
$$K = \frac{2.303 \times 0.3010}{t_{1/2}}$$

$$t_{1/2} = \frac{2.303 \times 0.3010}{K}$$

Unit: s

* GRAPHICAL REPRESENTATION [1st order reaction (half line)]

The Graph of concentration versus time is shown below. From the graph it is clear that ~~but~~ successive half life for first order reaction is constant.



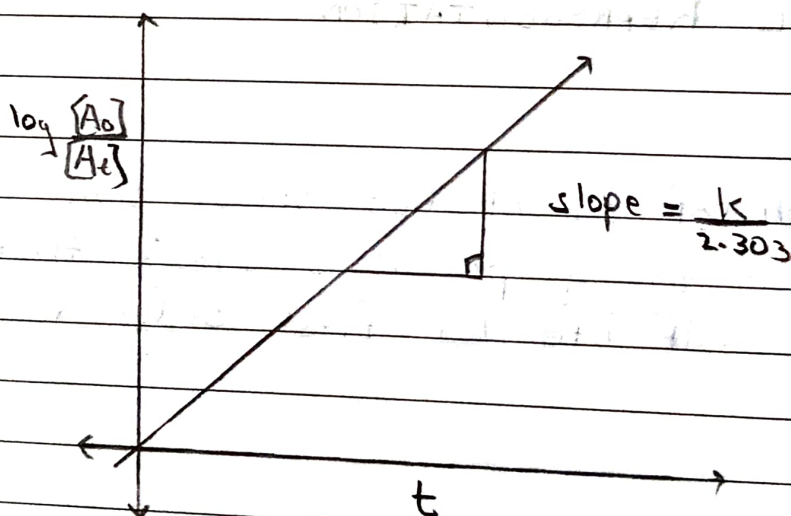
* GRAPHICAL REPRESENTATION [1st order Reaction]

I
$$kt = 2.303 \log \frac{[A_0]}{[A_t]}$$

$$\log \frac{[A_0]}{[A_t]} = \frac{k}{2.303} t + 0$$

which is of type
 $y = mx + c$

When graph of $\log \frac{[A_0]}{[A_t]}$ is plotted against t
we get a straight line passing through
origin with slope $\frac{k}{2.303}$



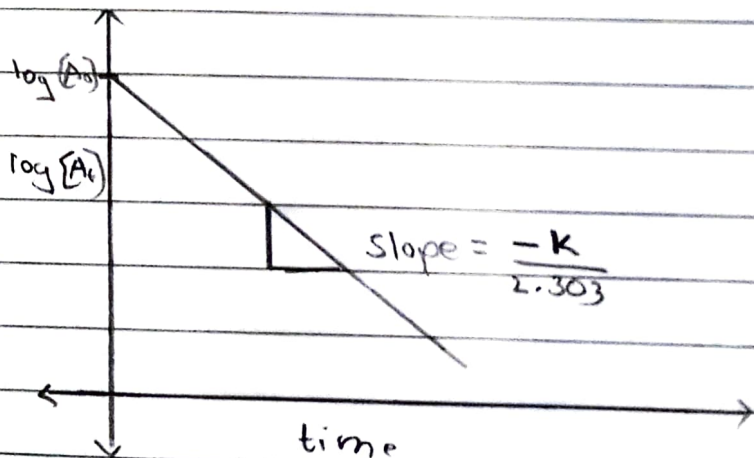
$$II \quad Kt = 2.303 \log \frac{[A_0]}{[A_t]}$$

$$\log [A_0] - \log [A_t] = \frac{Kt}{2.303}$$

$$-\log [A_t] = \frac{K}{2.303} t - \log [A_0]$$

$$\log [A_t] = -\frac{K}{2.303} t + \log [A_0]$$

Which is of type
 $y = mx + c$



$$III \quad \text{Rate} = -\frac{d[A]}{dt} = K[A]$$

which is of type $y = mx$

